Production Order List — CSV Export — API Specification Design (API仕様設計)

> **Revision of 2026-10-24 (WI-017).** This file supersedes `002_DD-API-CSV_製造指示一覧.md`, which stays unchanged as the earlier approved revision. Change: the production-order notes limit is 1000 characters (REQ-090, supersedes the 500 of REQ-016); see `work-items/WI-017/`.

002_DD-API-CSV — supports 002_DD-CSV, implements 002_BD-CSV FN-041/FN-042, requirements REQ-085–REQ-088.

Document control (改版履歴)

Field	Value
Document ID	002_DD-API-CSV
System name	ProductionManagementAI
Subsystem name	Production orders
Work item	WI-016
Created by	Claude (for ThongTM)
Created date	2026-10-07
Last updated by	Claude (for ThongTM)
Last updated date	2026-10-24

Version	Date	Author	Revision content
1	2026-10-07	Claude (for ThongTM)	Initial creation: API-PO-05; answers the 002_BD-CSV open questions on the row-limit status and the row-count header
2	2026-10-24	Claude (for ThongTM)	WI-017 (REQ-090–REQ-092): notes limit 500 → 1000 characters. Column 10 `備考` cites ≤ 1000 characters (001_BD revision 10). No contract change. Dated file name per WI-017 DEC-002

Overview and operation catalog

Field	Value
API / module name	Production orders API — CSV export
Overview	One read-only endpoint that returns, as a CSV file, every production order matching the list's filters in the list's sort order

No	Endpoint / method	Handler	Purpose	Notes
1	`GET /api/production-orders/export`	`ProductionOrdersController.Export` (002_DD-CSV module 6) → `ProductionOrderService.ExportAsync` (002_DD-FN-CSV)	CSV export of the filtered list	New

System-wide API registry entries

ID	Endpoint	Overview	Notes
API-PO-05	`GET /api/production-orders/export`	Export production orders matching the filters as CSV	New; continues the registry of 001_DD-API and 002_DD-API (API-PO-01–API-PO-04 unchanged)

Endpoint / method design

1. GET /api/production-orders/export

Field	Value
Description	Return all production orders matching the filters, in the requested order, as one UTF-8 CSV file with a header row; refuse when more than 10,000 orders match
Return type	File (`text/csv`), streamed
Created by / date	Claude / 2026-10-07
Last modified by / date	—

Arguments: query-string parameters only — see Request fields. No route or body parameters.

Processing overview: authorize with the list's policy; validate the query string with the list's validator (`ProductionOrderListQuery.TryCreate` , 002_DD-API §1 rules, paging left at its defaults and not used); verify the product filter; in one read-only transaction count the matches, refuse with 422 when the count exceeds 10,000,

otherwise send the response headers (including the count) and stream the header row and one CSV row per order in the requested order. Formatting and escaping are FN-042 (002_DD-FN-CSV).

Processing flow

Step	Description	Calls
1	Authorize (`Admin` / `Operator`)	policy `ProductionOrderEditor` (class attribute, as the list)
2	Validate the query; on failure 400 with per-parameter message IDs	`ProductionOrderListQuery.TryCreate` (unchanged)
3	Verify the product filter exists, when sent; else 400 `productId: MSG-E002`	`IProductionOrderRepository.ProductExistsAsync` (existing)
4	Open a read-only snapshot; count matches	`ProductionOrderService.ExportAsync` → 002_DD-FN-CSV §1
5	Count > 10,000 → 422 `MSG-E024`	002_DD-FN-CSV §1
6	200: headers, then BOM, header row and rows, streamed	002_DD-FN-CSV §2, §3

Return value

Type	Name	Description
CSV file	body	See "Response body (CSV)" below

Request fields

The filter and sort parameters of API-PO-04 (002_DD-API §1 request fields 1–7), with the same names, types, defaults, allow-lists and error message IDs. They are not restated here.

No	Name	Variable name	Type	Length	Required	Source	Example	Description	Notes
1–7	Status, Product, Due from, Due to, Order number, Sort, Direction	`status` (repeatable), `productId` , `dueFrom` , `dueTo` , `orderNumber` , `sort` , `dir`	as 002_DD-API	as 002_DD-API	no	query	`?status=InProgress&status=Draft&dueFrom=2026-10-01&dueTo=2026-10-31&sort=dueDate&dir=asc`	Same meaning as on the list	Same validation and message IDs (002_BD V-09–V-13); default sort `dueDate` `asc` , order number ascending as tie-breaker

No	Name	Variable name	Type	Length	Required	Source	Example	Description	Notes
—	Page, Page size	`page`, `pageSize`	—	—	—	—	—	Not parameters of this endpoint	Ignored if sent, like any unknown query parameter; the export always covers all pages (DEC-002). The client does not send them (002_DD-CSV module 3)

Response headers (200)

No	Header	Value	Example	Notes
1	`Content-Type`	`text/csv; charset=utf-8`		REQ-087
2	`Content-Disposition`	`attachment; filename="production-orders_{yyyyMMdd-HHmm}.csv"; filename*=UTF-8''{percent-encoded 製造指示一覧_{yyyyMMdd-HHmm}.csv}`	`attachment; filename="production-orders_20261007-1345.csv"; filename*=UTF-8''%E8%A3%BD%E9%80%A0%E6%8C%87%E7%A4%BA%E4%B8%80%E8%A6%A7_20261007-1345.csv`	002_BD-CSV M-16; time is plant-local (`IPlantClock`) at the start of the export (RFC 6266 / RFC 8187)
3	`X-Total-Count`	Number of data rows in the body	`87`	Decided here (BD open question). Equals the count from step 4, which reads the same snapshot as the rows (002_DD-FN-CSV §1). Read by the client for MSG-I009
4	`Cache-Control`	`no-store`		Existing class-level `[ResponseCache(NoStore = true)]`

No `Content-Length`: the body is streamed (chunked). Same origin, so no CORS exposure of `X-Total-Count` is needed.

Response body (CSV)

Encoding UTF-8 with BOM (`EF BB BF`) as the first three bytes; lines end with CRLF, including the last; separator comma; RFC 4180 quoting; formula-like text neutralised (002_BD-CSV CSV file layout, DEC-004). The header row is always present, so a body has 1 + `X-Total-Count` lines.

No	Column header	Source	Type in file	Value mapping	Example	Notes
1	指示番号	`production_orders.order_number`	text	—	`PO-2026-00041`	

No	Column header	Source	Type in file	Value mapping	Example	Notes
2	`製品コード`	`products.sku`	text	—	`P-1001`	
3	`製品名`	`products.name`	text	—	`ブレーキキャリパー`	Neutralised if formula-like
4	`生産ライン`	`production_lines.code` , `.name` , `.is_active`	text	`{code} – {name}` , plus `（使用停止）` if retired; empty if none (M-14)	`L-01 – 第1組立ライン`	
5	`数量`	`production_orders.quantity`	number	Invariant culture, `.` decimal point, no grouping, trailing zeros removed (M-12)	`12.5`	Never neutralised
6	`単位`	`products.unit`	text	—	`kg`	
7	`納期`	`production_orders.due_date`	date	`yyyy/MM/dd` (M-13)	`2026/10/02`	Plant-local date, no conversion
8	`ステータス`	`production_orders.status`	text	`Draft` → `下書き` , `InProgress` → `進行中` , `Completed` → `完了` , `Cancelled` → `取消` (M-11)	`進行中`	
9	`納期遅れ`	computed	text	`納期遅れ` when due date < plant today and status ∈ {Draft, InProgress}, else empty (M-15)	`納期遅れ`	Same rule as `isOverdue` in API-PO-04
10	`備考`	`production_orders.notes`	text	As stored; null → empty; quoted when it contains `,` `"` CR or LF	`"梱包は""A""仕様、2段積み"`	≤ 1000 characters (001_BD revision 10, WI-017); neutralised if formula-like
11	`作成日時`	`production_orders.created_at_utc`	timestamp	UTC → `Asia/Tokyo` , `yyyy/MM/dd HH:mm` (M-13, DEC-008)	`2026/09/28 09:15`	
12	`更新日時`	`production_orders.updated_at_utc`	timestamp	as column 11	`2026/09/30 14:02`	
13	`完了日時`	`production_orders.completed_at_utc`	timestamp	as column 11; empty when null	`2026/10/05 17:30`	Set by WI-004 on `InProgress → Completed`

Header row (exact): `指示番号,製品コード,製品名,生産ライン,数量,単位,納期,ステータス,納期遅れ,備考,作成日時,更新日時,完了日時`

Sample file (BOM, CRLF, all edge cases: decimal quantity, retired line, unassigned line, notes with comma, quotes, line break and a leading `=`): [`mockups/002_DD-API-CSV_sample.csv`](mockups/002_DD-API-CSV_sample.csv) . Excerpt (BOM not shown):

```
指示番号,製品コード,製品名,生産ライン,数量,単位,納期,ステータス,納期遅れ,備考,作成日時,更新日時,完了日時
PO-2026-00041,P-1001,ブレーキキャリパー,L-01 – 第1組立ライン,120,個,2026/10/02,進行中,納期遅れ,,2026/09/20 10:12,2026/09/28 09:15,
PO-2026-00052,P-2002,防錆塗料,L-03 – 塗装ライン（使用停止),12.5,kg,2026/10/03,下書き,,"梱包は""A""仕様、2段積み",2026/09/29 08:40,2026/09/30 14:02,
PO-2026-00060,P-1010,エアフィルター,,300,個,2026/10/12,下書き,,'=1+1 は顧客指定の品番表記,2026/10/01 13:00,2026/10/01 13:00,
```

Error codes

Errors are RFC 9457 Problem Details (`application/problem+json`) with `code` and `traceId`, exactly as the other production-order endpoints (`ProductionOrderProblems`); no CSV body is sent with an error.

Code	Meaning	HTTP status
`VALIDATION` with `errors` `MSG-E015`–`MSG-E019`, `MSG-E002`	Invalid filter or sort parameter (same keys and IDs as API-PO-04)	400
—	Not signed in	401
—	Missing role (client shows MSG-E020)	403
`MSG-E024` (type `urn:pmai:problem:rule-violation`)	More than 10,000 orders match; decided here (BD open question): the existing `Result.RuleViolation` → 422 mapping, so no new error shape	422
`MSG-E013`	Unexpected failure before the first byte of the body	500
—	Failure after streaming started: the connection is aborted, so the client sees a network error, never a truncated "successful" file (002_DD-FN-CSV §3)	(connection reset)

Example 422 body:

```
{
  "type": "urn:pmai:problem:rule-violation",
  "title": "The change breaks a production order rule.",
  "status": 422,
  "code": "MSG-E024",
  "traceId": "00-4bf92f3577b34da6a3ce929d0e0e4736-00f067aa0ba902b7-01"
}
```

The `title` is the shared English title of the existing rule-violation mapping; it is API text and never shown in the UI (the client shows MSG-E024 from its catalog).

Unresolved decisions

None. Decided in this document: 422 `MSG-E024` for the row limit and `X-Total-Count` for the row count (both 002_BD-CSV open questions). The policy limitation of API-PO-04 applies unchanged: the endpoint is gated by `ProductionOrderEditor` until the role/permission matrix (WI-001 DEC-015) is settled.